Microbial community / environmental DNA analysis with Chipster (Day 2)

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Outline of Day 1: Preprocessing reads

Quality check

↓

(Remove primers)

↓

Filter and trim reads

↓

Sample inference ⟶ ASVs

↓

Create contigs

↓

Remove chimera

↓

Assign taxonomt

↓

Create a phyloseq object

Output so far: a phyloseq object with

1. ASV table (counts in each sample)
2. taxonomy of ASVs
3. sample information (phenodata)
4. sequences of ASVs

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Day 2: Community analysis

- Data tidying & transformations
- Taxonomy plots
- Alpha diversity
- Beta diversity: ordinations & statistics

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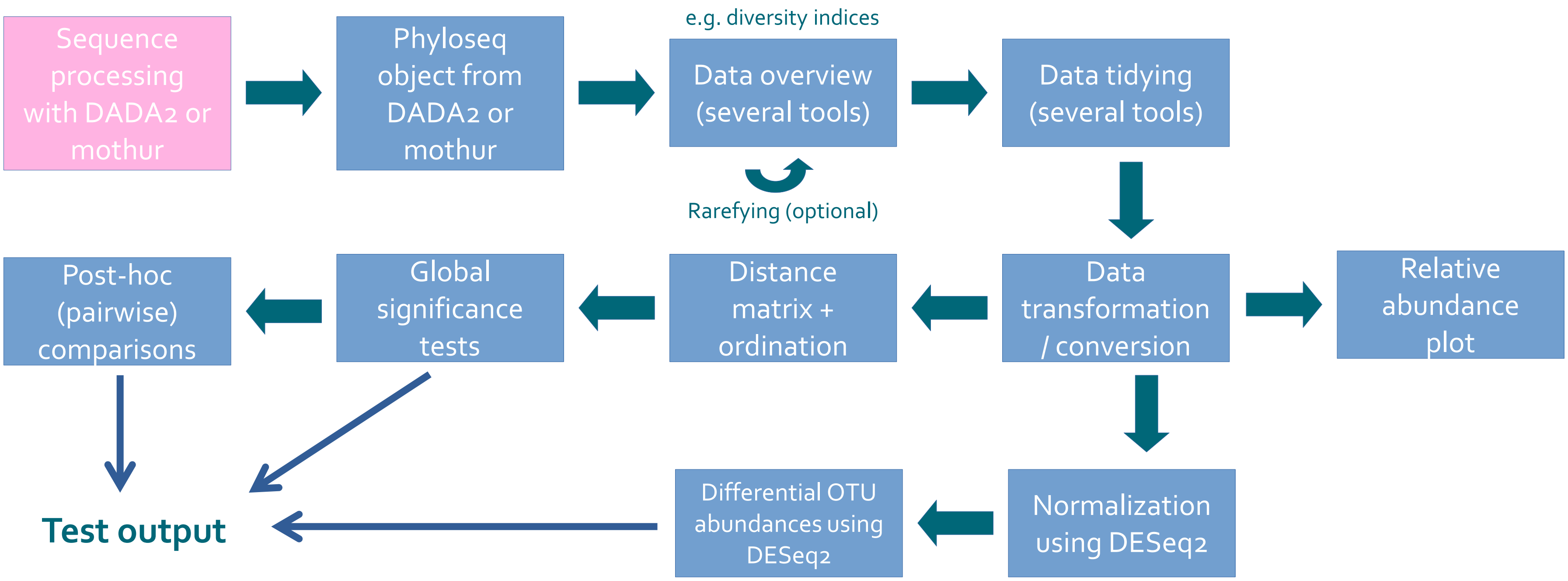
Part 1: Tool overview

Workflow for community analysis in Chipster



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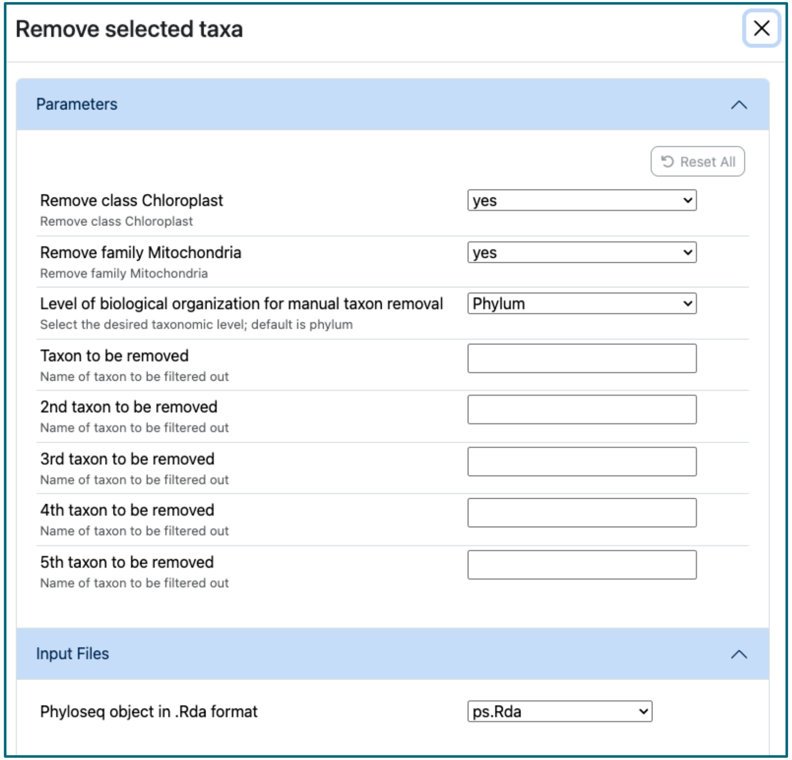
Part 2: Data tidying and alpha diversity

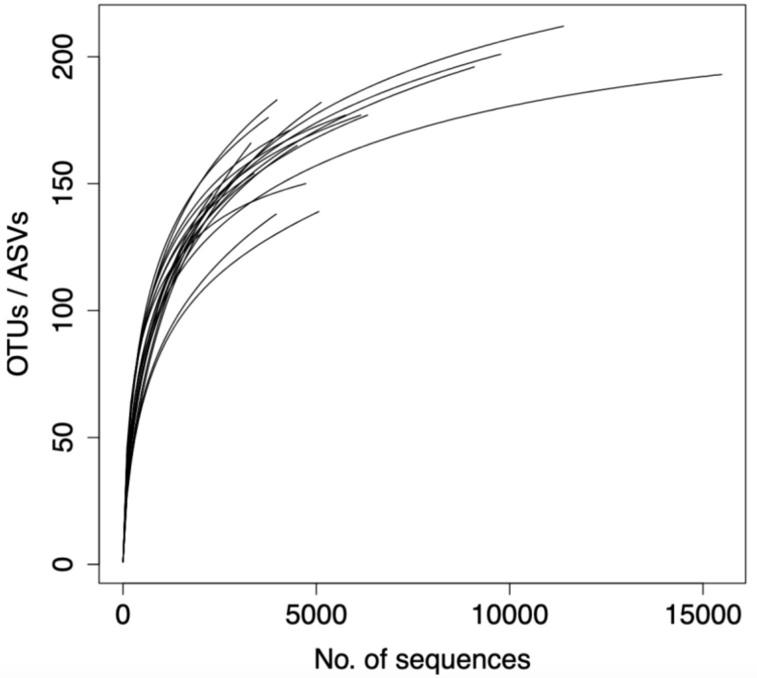
Taxon-level clean-up tools

Under Microbial amplicon data analyses:

- Filter by taxonomic group
  - Remove non-specific sequences (keep e.g. Bacteria or Archaea only)
- Remove selected taxa
  - Remove chloroplast and/or mitochondrial sequences
  - (Manually remove specific taxa)
- Overview of taxon composition
  - user-specified level



Data inspection: Sequence numbers, rarefaction curve and alpha diversity estimates




```
### Per-sample sequence no.s ###

  F3D0   F3D1 F3D141 F3D142 F3D143 F3D144 F3D145 F3D146 F3D147 F3D148 F3D149
  6149   4727   4504   2311   2347   3307   5125   3756  11390   9085   9770
F3D150   F3D2   F3D3   F3D5   F3D6   F3D7   F3D8   F3D9
  3981  15481   5062   3387   6323   3962   4306   5758

### Alpha diversity estimates (observed OTUs, Chao1, Shannon's index, Pielou's evenness) ###

       Observed    Chao1  se.chao1  Shannon    pielou  time
F3D0        177 210.0000 15.261725 3.853259 0.7444256 early
F3D1        150 163.1538  7.993669 4.032404 0.8047691 early
F3D141      165 194.6897 12.042909 3.499614 0.6853998  late
F3D142      146 193.0400 17.787187 3.343976 0.6709952  late
F3D143      143 177.7308 13.935854 3.481141 0.7014407  late
F3D144      166 246.6400 27.261107 3.212135 0.6283534  late
F3D145      182 231.2143 17.858418 3.160095 0.6072427  late
F3D146      176 219.1579 17.951815 3.758571 0.7269283  late
```

Filtering low-abundance ASVs

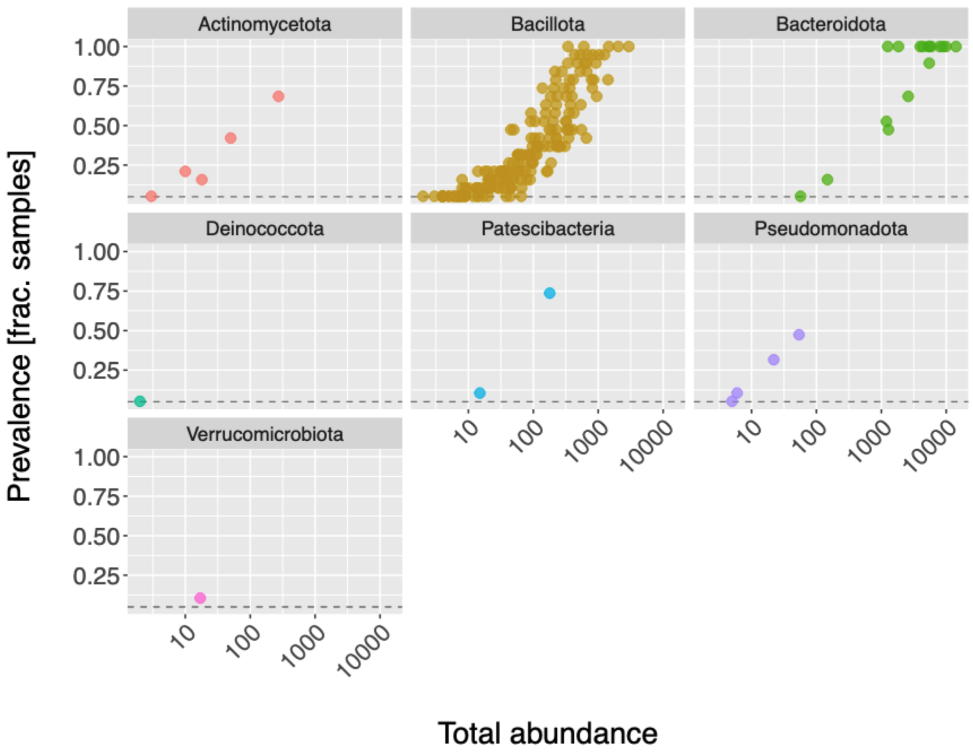
- Filter out ASVs that occur in less than x % of samples
    - Proportional prevalence filtering

- Remove ~~singletons and~~ doubletons
    - Remove OTUs with specific number of occurrences

Visualizing ASV counts

Prevalence summaries

- Plot of ASV counts (x axis) vs. fraction of samples they occur in (y axis) at phylum level
  - Each dot = ASV
- Text summary on low prevalence and low abundance ASVs



Alpha diversity

Diversity within a habitat: how many species are present, at which relative amounts?

- Species richness - how many species?
  - observed number of ASVs
  - richness estimators (Chao1)
- Evenness – abundance distribution of species?
  - all ASVs equally abundant vs. a few dominant and a lot of rare ASVs
  - Pielou's evenness
- Many diversity indices **combine richness and evenness** (e.g. Shannon index)

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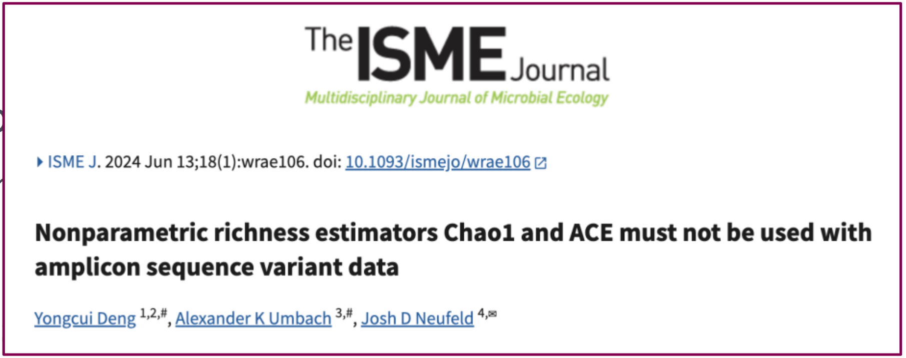
note: do not use after removing <u>singletons</u> and doubletons!

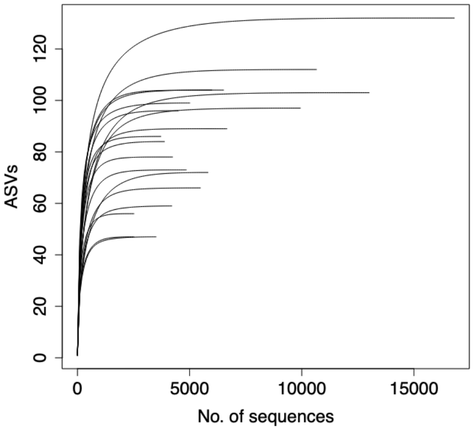

The ISME Journal
Multidisciplinary Journal of Microbial Ecology

ISME J. 2024 Jun 13;18(1):wrae106. doi: 10.1093/ismejo/wrae106

Nonparametric richness estimators Chao1 and ACE must not be used with amplicon sequence variant data

Yongcui Deng [1,2,#], Alexander K Umbach [3,#], Josh D Neufeld [4,✉]

Rarefaction?



Uneven sequence numbers among samples can bias comparisons, especially with **alpha diversity**. Solutions:

- rarefying: equal number of reads picked from all samples (Rarefy ASV data to even depth)
- data transformation (Transform ASV counts)

PLOS COMPUTATIONAL BIOLOGY

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RESEARCH ARTICLE

Waste Not, Want Not: Why Rarefying Microbiome Data Is Inadmissible

Paul J. McMurdie, Susan Holmes

Published: April 3, 2014 • https://doi.org/10.1371/journal.pcbi.1003531

| Editor's Pick | Microbial Ecology | Research Article | 6 December 2023

Waste not, want not: revisiting the analysis that called into question the practice of rarefaction

Author: Patrick D. Schloss | AUTHORS INFO & AFFILIATIONS

https://doi.org/10.1128/msphere.00355-23 • Check for updates

| Editor's Pick | Human Microbiome | Research Article | 22 January 2024

Rarefaction is currently the best approach to control for uneven sequencing effort in amplicon sequence analyses

Author: Patrick D. Schloss | AUTHORS INFO & AFFILIATIONS

https://doi.org/10.1128/msphere.00354-23 • Check for updates

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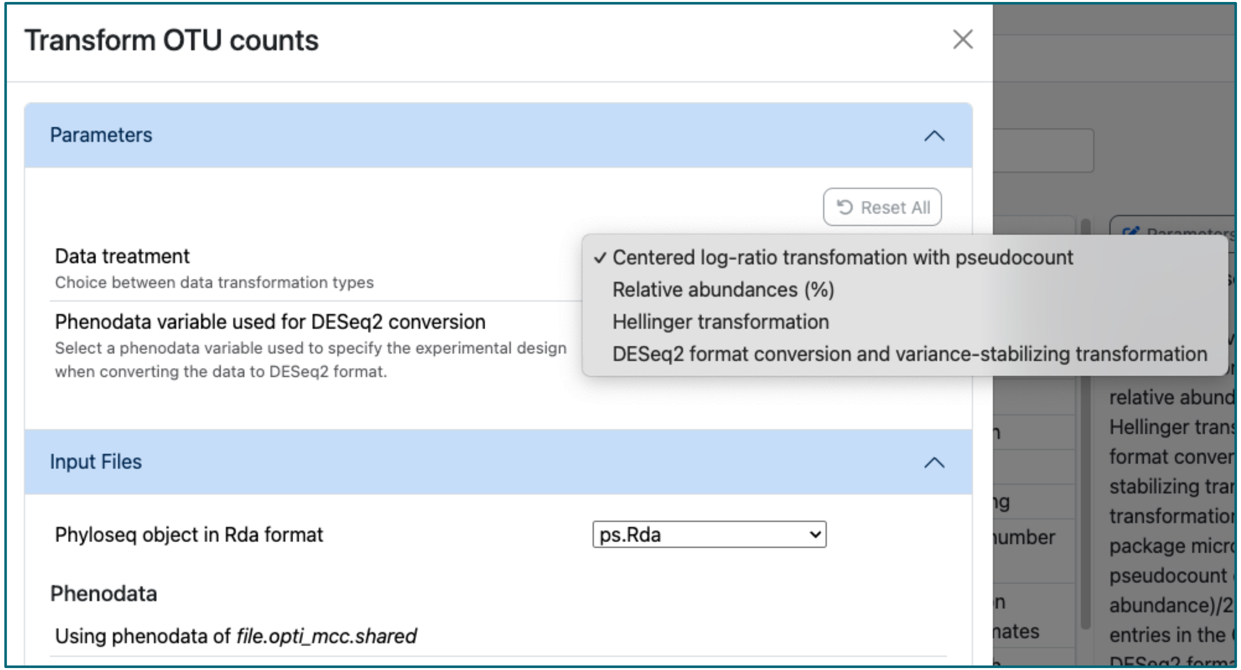
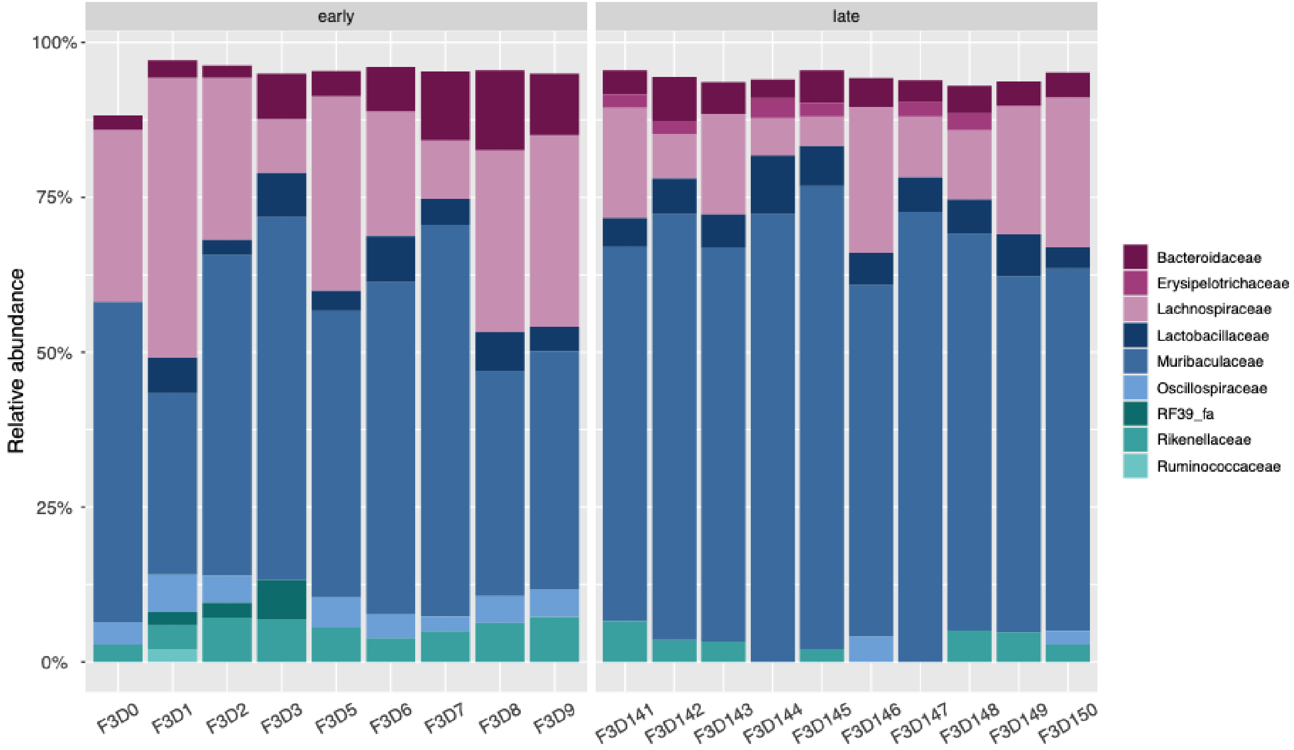
Part 3: Transformations and ordinations

Transformation of ASV data

Four options



Relative abundance (%) bar plots



Beta diversity

Change in community composition among habitats

- Does the microbial community composition in treatment A differ from treatment B?

- Unlike in alpha diversity, ASV identity matters
- Both identity and relative abundance of ASVs usually included
- Quantified with **distance or dissimilarity measures**

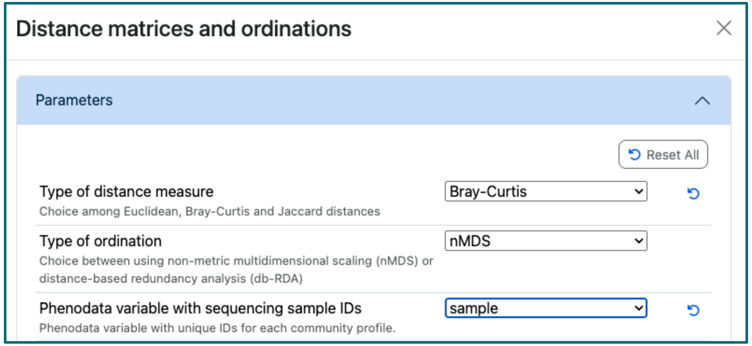
Distance matrices and ordinations

Distance measures available: **Bray-Curtis** or Euclidean

- centered log ratio (CLR) transformation + Euclidean distance = **Aitchinson distance**

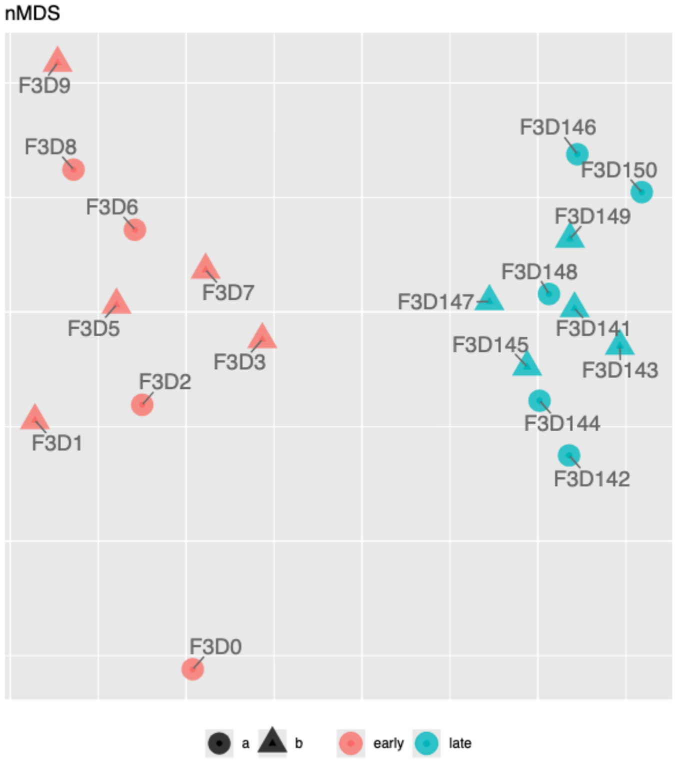
Ordinations: visualizing beta diversity

- nMDS (non-metric multidimensional scaling)
    - overall variation among samples displayed
- db-RDA (distance-based redundancy analysis)
    - focus on the variation explained by phenodata variable(s)



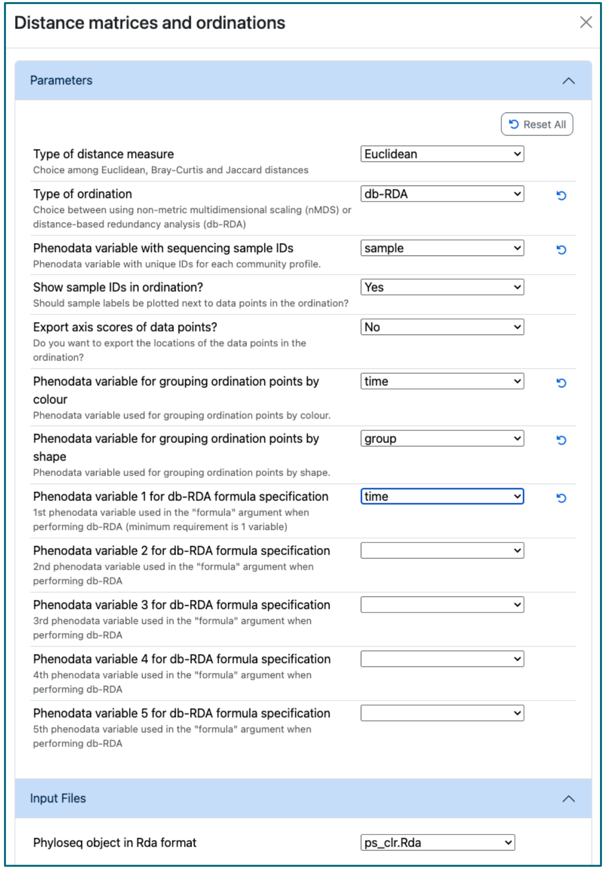
Recommended for more information:
Guide to Statistical Analysis in Microbial Ecology:
https://sites.google.com/site/mb3gustame/

Non-metric multidimensional scaling (nMDS)



Distance-based redundancy analysis (db-RDA)

Requires specifying one or more phenodata variables



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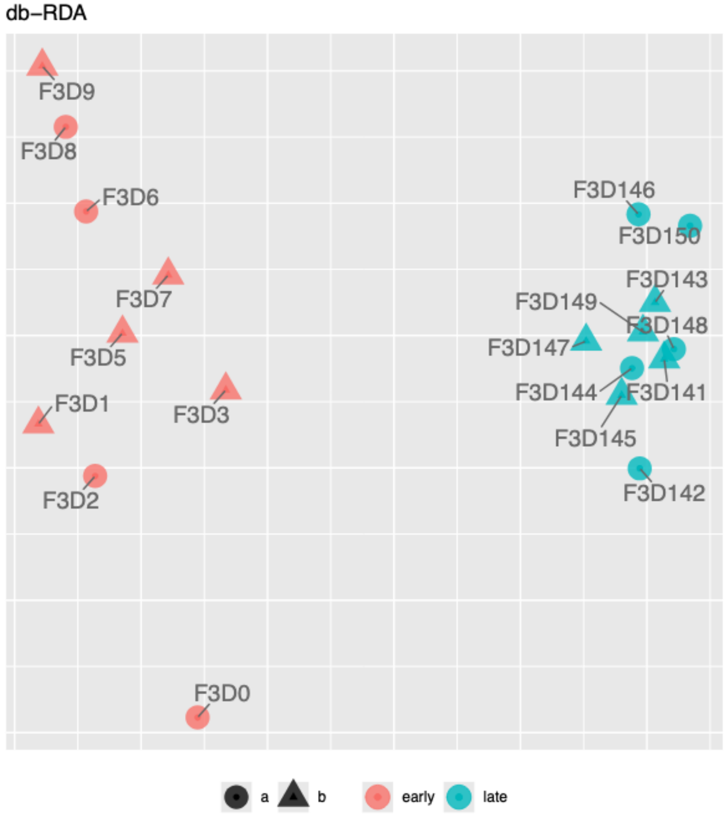

Constrained ordination = focus on the community variation explained by the phenodata variable(s)

Recommended for more information:
Guide to Statistical Analysis in Microbial Ecology: https://sites.google.com/site/mb3gustame/

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Part 4: Statistics

PERMANOVA (permutational multivariate analysis of variance)

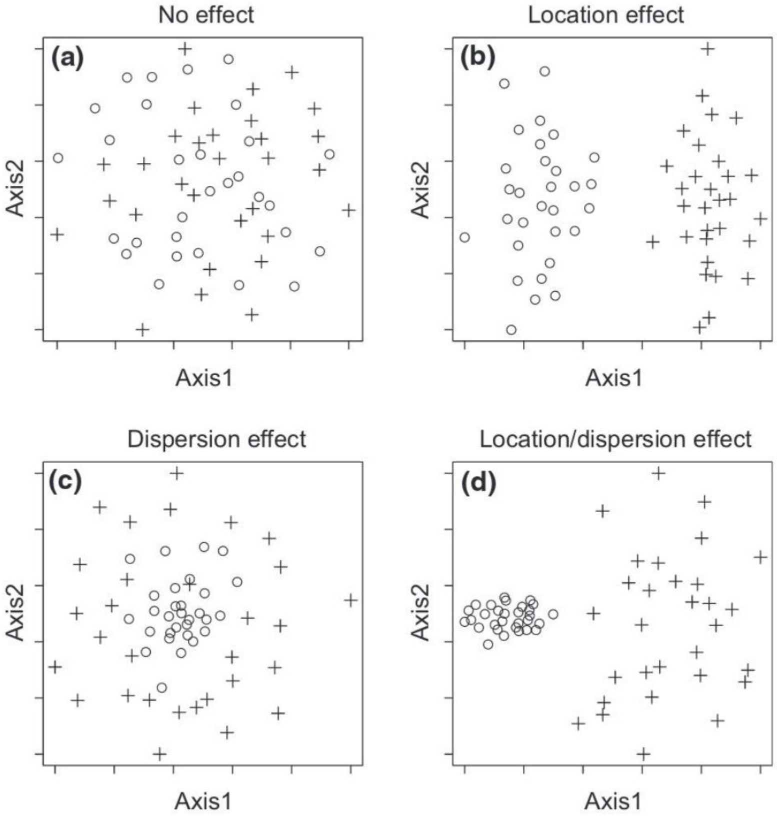
Input: distance matrix (ps_dist.Rda)

- Global test: '**Does community structure differ between sample groups**?'
    - Pairwise test: 'Which groups differ from one another?'
- Currently: several phenodata variables -> added sequentially -> order matters!
- Influenced by both **location** and **dispersion**

Location vs. dispersion

A significant PERMANOVA result can be due to:

- Location effect
- Dispersion effect
- Combination of both



Source: https://doi.org/10.1111/j.2041-210X.2011.00127.x

PERMANOVA output

```
### Global PERMANOVA summary ###


Permutation test for adonis under reduced model
Marginal effects of terms
Permutation: free
Number of permutations: 999

adonis2(formula = ps_dist ~ get(pheno1), data = ps_df, by = "margin")
            Df SumOfSqs      R2      F Pr(>F)    
get(pheno1)  1   2228.1 0.32901 8.3359  0.001 ***
Residual    17   4544.0 0.67099                  
Total       18   6772.1 1.00000                  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Df: Degrees of freedom

F.Model: Test statistic (pseudo-*F*)

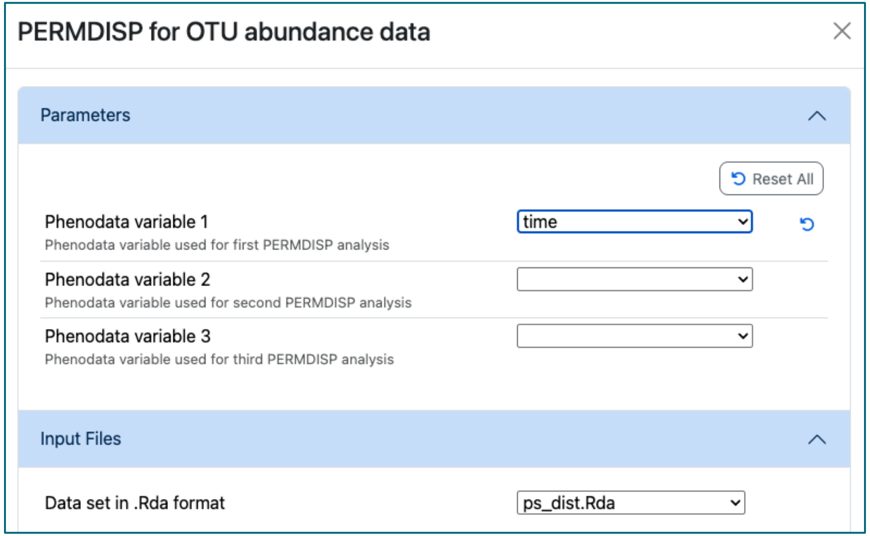
R2: Variation explained by the model

Pr(>F): Statistical significance (*p* value)

PERMDISP: test for the homogeneity of multivariate dispersion

Input: distance matrix

- Tests if a significant PERMANOVA result is due to dispersion, not (only) location
- Significant result → be careful with PERMANOVA interpretation



Post-hoc comparisons

Sample group comparisons **following significant global PERMANOVA:**

- Pairwise PERMANOVA (similar as global test but for sample pairs)

Dispersion comparisons **following significant PERMDISP:**

- Tukey’s Honestly Significant Difference (HSD) test

- Both methods use a correction for multiple testing (Benjamin-Hochberg correction)

DESeq2

- Originates from the RNA-seq field
- Addresses the question: '**Which taxa are differentially abundant between sample groups?'**
- Enables inferences such as: 'Illness x is associated with a reduction in the abundance of beneficial gut microbes y and z'
- Input: untransformed data → convert to DESeq2 format with Transform OTU counts (corrects for differences in sequencing depth)
- Results given as **log fold changes**

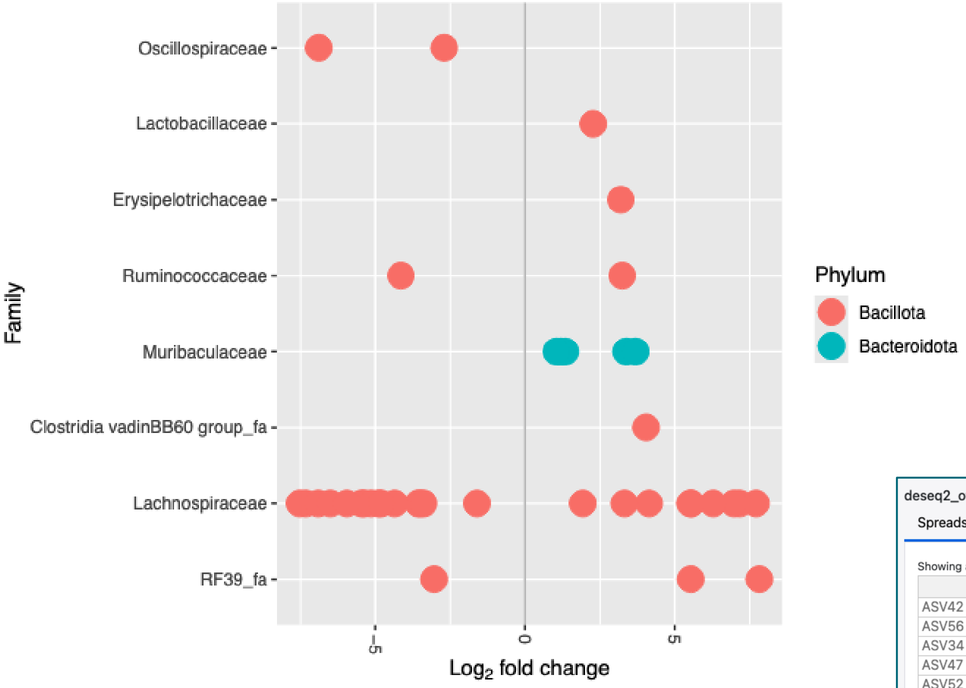
More information: https://joey711.github.io/phyloseq-extensions/DESeq2.html

DESeq2

Current tool configuration:

- **Focused on comparison of two groups at a time**
    - If selected phenodata column has >2 groups, specify a pair (Group 1 and Group 2)
- **Reference level selection**:
    - Phenodata column with two groups -> first in alphabet is the reference level (e.g. 'b vs. a' or 'sick vs. healthy')
    - Phenodata column with >2 groups -> 'Group 2' is the reference level

DESeq2



8-fold increase compared to reference level = log2 fold change 3 (because 2^3=8)

each dot = OTU with adjusted p value < 0.01

deseq2_otutable.tsv ...

Spreadsheet | Text | Volcano Plot | Open in New Tab | Details

Showing all 39 rows. Full Screen

	baseMean	log2FoldChange	lfcSE	padj	Phylum	Class	Order
ASV42	16.4236259611121	-7.31840767802812	0.749154137467773	2.0364125932923e-20	Bacillota	Clostridia	Lachnospirales
ASV56	18.3391701526785	7.81985204520067	0.873455049412247	2.30513020461074e-17	Bacillota	Bacilli	RF39
ASV34	18.8618023780781	-7.52571922193085	0.866068695314826	1.61365138573453e-16	Bacillota	Clostridia	Lachnospirales
ASV47	12.2503485132916	-6.90746701234306	0.824554823282435	1.80208867026377e-15	Bacillota	Clostridia	Lachnospirales
ASV52	16.7848262358729	7.70808179943063	0.926727682754117	2.3891649193975e-15	Bacillota	Clostridia	Lachnospirales
ASV57	12.1431457248744	-6.88344590854819	1.24751573627951	7.6124929030845e-07	Bacillota	Clostridia	Oscillospirales
ASV61	9.27437283192242	-6.49968931118565	1.19304098226415	9.39990693753832e-07	Bacillota	Clostridia	Lachnospirales
ASV63	11.4810124085797	7.16117160669036	1.31895955099439	9.39990693753832e-07	Bacillota	Clostridia	Lachnospirales